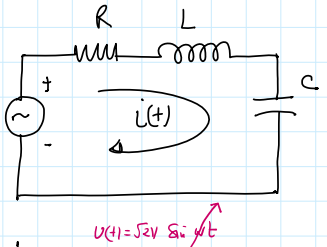
Lecture 20: Resonance, Frequency Response and Filters

Resonance: A resonance is a common phenomenon in physical world. A resonance happens when there are two independent energy elements in electrical, mechanical, pneumatic etc systems.

For RLC series circuit, resonance happens when,

$$\bar{I} = \frac{\bar{V}}{R + j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{\bar{V}}{Z}$$



When resonance takes place, the current in phase with voltage. This happens, when

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}; \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \text{OR} \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

ω_0 = resonance frequency

At resonance $Z = R$; $\bar{I} = \frac{\bar{V}}{R} = \frac{V}{R}\angle 0 \text{ A} = I_0$

⇒ The current at resonance is in phase with the voltage, $\bar{V}$.

Nature of Z: (A) If $\omega < \omega_0$,

$\frac{1}{\omega C} > \omega L$, Z is capacitive, $Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$

(B) If $\omega > \omega_0$, Z is inductive

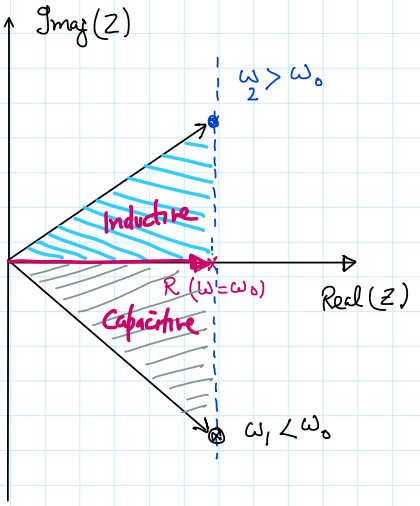
$\omega L > \frac{1}{\omega C}$; Z is inductive, $Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$

(C) If $\omega = \omega_0$

$Z = R$, Z is resistive

$\omega_1 < \omega_0 < \omega_2$

Capacitive → Resistive → Inductive



Frequency Response:

Depending upon the frequency of excitation ω, and resonance frequency, ω_0, the current response will vary, as given following.

In general, $\bar{I} = \dfrac{\bar{V}}{R + j\left(\omega L - \frac{1}{\omega C}\right)} = \dfrac{V/R}{1 + j\left(\frac{\omega L}{R} - \frac{1}{\omega C R}\right)}$

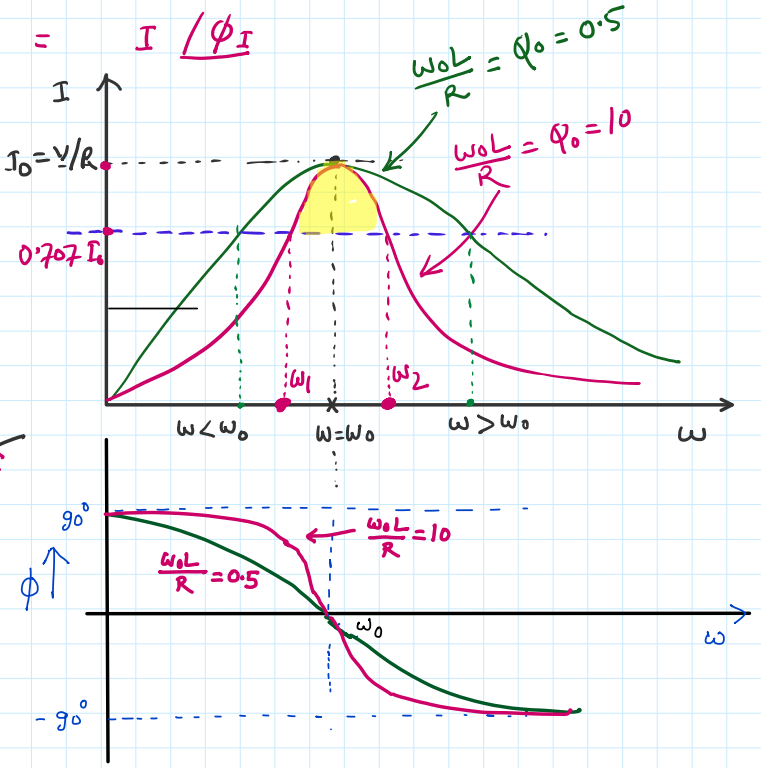
$$= \frac{V/R}{1 + j\left(\frac{\omega_0 L}{R}\frac{\omega}{\omega_0} - \frac{1}{\omega_0 C R}\frac{\omega_0}{\omega}\right)} \qquad \omega_0 L = \frac{1}{\omega_0 C}$$

$$= \frac{V/R}{1 + j\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\bar{I} = \left\{\frac{V/R}{\sqrt{1 + \left(\frac{\omega_0 L}{R}\right)^2\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}\right\} \angle - \tan^{-1}\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)$$

Bandwidth, ω_b is defined range of frequencies for which the power delivered to the Load is half of the power at resonance.

$= I \angle \phi_I$



$P_0 = I_0^2 * R$

$P = \frac{P_0}{2} = I_1^2 R = \frac{I_0^2 R}{2}$

$I_1 = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$

$\omega_b = \omega_2 - \omega_1$

$$I = \left|\frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)}\right| = \frac{I_0}{\sqrt{2}} = \frac{V}{\sqrt{2}R}$$

$$\sqrt{2}R = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$2R^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 \Rightarrow R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$\pm R = \left(\omega L - \frac{1}{\omega C}\right)$$

① Taking +R

$$\pm R = \left(\omega L - \frac{1}{\omega C}\right)$$

(A) Taking $+R$

$$R = \omega L - \frac{1}{\omega C}$$

$$R = \frac{\omega^2 LC - 1}{\omega C}$$

$$\omega^2 LC - RC\omega - 1 = 0$$

$$\omega^2 - \frac{R}{L}\omega - \frac{1}{LC} = 0$$

$$\omega = \frac{\frac{R}{L} \pm \sqrt{(R/L)^2 + \frac{4}{LC}}}{2} = \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2, \omega_2' = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \; ; \; \frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

= higher cut-off frequency

(B) $-R = \left(\omega L - \frac{1}{\omega C}\right) \Rightarrow$

$$\omega^2 + \frac{R}{L}\omega - \frac{1}{LC} = 0$$

$$\omega_1, \omega_1' = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}}{2}$$

$$\omega_1, \omega_1' = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \; ; \; -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

= lower cut-off frequency

$$\boxed{\omega_b = \omega_2 - \omega_1 = \frac{R}{L}}$$

$$Q_0 = \frac{\omega_0}{\omega_b} = \frac{1 * L}{\sqrt{LC}\, R} = \frac{\sqrt{L\omega_0}}{\sqrt{C\omega_0} \times R} = \left(\frac{\omega_0 L}{R}\right) = \frac{X_{L0}}{R}$$

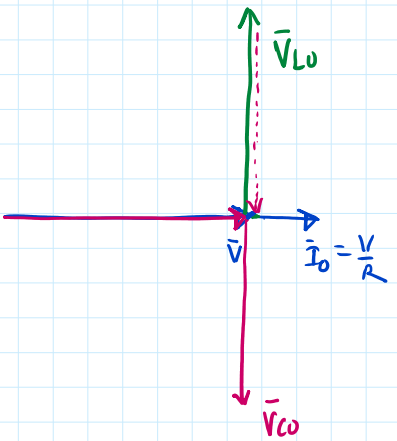
Voltages:

$$\bar{V}_{L0} = I_0 * j\omega_0 L =$$

$$= I_0 * \omega_0 L \,\angle 90^\circ$$

$$= \frac{V}{R}\omega_0 L \,\angle 90^\circ$$

$$\boxed{\bar{V}_{L0} = V Q_0 \,\angle 90^\circ}$$

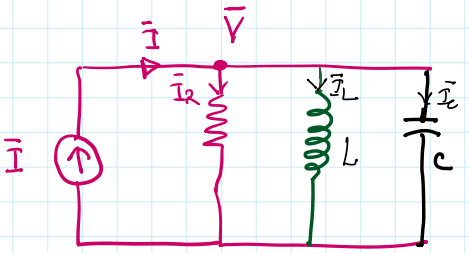


$$\bar{V}_{C0} = \bar{I}_0 * (-j/\omega_0 C)$$

$$\bar{V}_{C0} = V Q_0 \,\angle -90^\circ = -\bar{V}_{L0}$$

Resonance in RLC parallel Ckt:

$\bar{V} = \bar{I} * Z \Rightarrow \bar{I} = \bar{V} Y$



$$Y = Y_1 + Y_2 + Y_3$$
$$= \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$$
$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right) \Rightarrow \text{At resonance } \omega_0 C = \frac{1}{\omega_0 L}$$
$$= G + jB;\ G = \frac{1}{R};\ B = \omega C - \frac{1}{\omega L}$$

At resonance: $P_0 = \frac{V_0^2}{R}$; and $\bar{V}_0 = \bar{I} * R$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

or $\bar{I} = \frac{\bar{V}_0}{R}$; $\bar{I}$ is current source and fixed

Half power of resonance power $= \frac{P_0}{2} = \frac{V_0^2}{2R} = \frac{(V_0/\sqrt{2})^2}{R} = \frac{V_{1/2}^2}{R}$

where $V_{1/2}$ is the voltage at half resonance power.

This gives $V_{1/2} = \frac{V_0}{\sqrt{2}}$

Now this voltage $V_{1/2}$ can be given at frequency ω;

$$\bar{I} = \bar{V}_{1/2}\left\{\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right\}$$

$$\frac{\bar{V}_0}{R} = \left|\bar{V}_{1/2} * \left\{\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right\}\right|$$

$$\frac{V_0}{R} = \frac{V_0}{\sqrt{2}}\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

$$\frac{\sqrt{2}}{R} = \frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2 \Rightarrow \frac{2}{R^2} = \frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2$$

$$\Rightarrow \frac{1}{R^2} = \left(\omega C - \frac{1}{\omega L}\right)^2$$

or $\omega C - \frac{1}{\omega L} = \pm\frac{1}{R}$

or $\omega^2 \pm \frac{1}{RC}\omega - \frac{1}{LC} = 0$

$$\Rightarrow \omega_{1,2} = \frac{\pm\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 + 4/LC}}{2}$$

$$\omega_{1,2} = \pm\frac{1}{2RC} \pm\sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\boxed{\omega_b = \omega_2 - \omega_1 = \frac{1}{RC}}$$

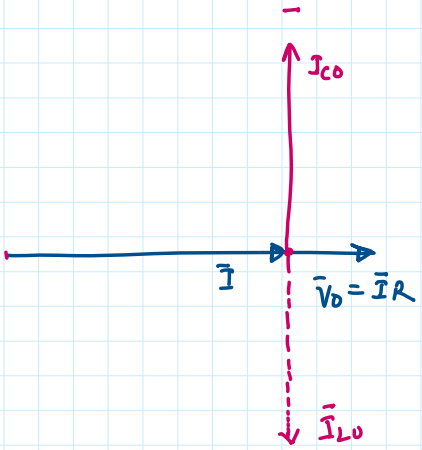
With this: $\omega_1 \approx \omega_0 - \frac{\omega_b}{2}$ | $\omega_0 = \frac{1}{\sqrt{LC}}$

$\omega_2 \approx \omega_0 + \frac{\omega_b}{2}$ | $\omega_b = \frac{1}{RC}$

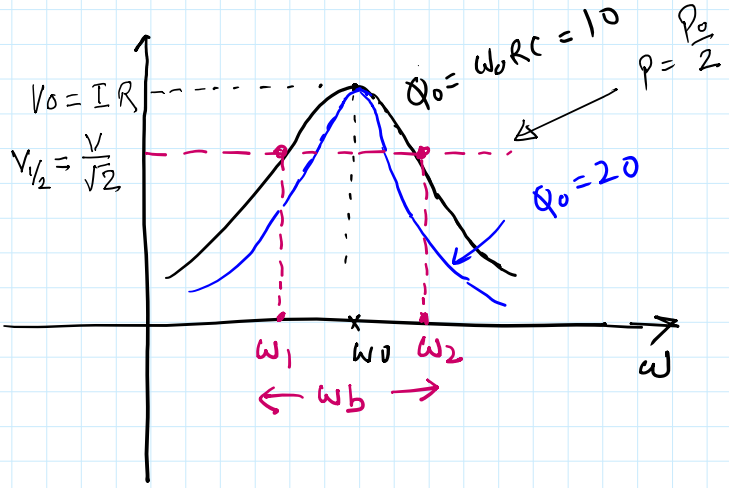
The above approximation holds true for narrower band i.e. $\omega_0 >> \frac{\omega_b}{2}$

Quality factor $Q_0 = \frac{\omega_0}{\omega_b} = \frac{\omega_0}{\frac{1}{RC}} = \omega_0 RC$

$$\boxed{Q_0 = \omega_0 RC}$$



Phasor Diagram



Resonance in RLC parallel circuit